

ACT vs. Diffusion Policy on a Home-Built Delta Robot: Data and Training-Length Scaling for Real-World Pick-and-Place

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Code, CAD, firmware, data and models: <https://github.com/KirkEHaskell/delta-robot-act-vs-diffusion>

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Abstract

We compare the two most widely used open-source imitation-learning policies, ACT and Diffusion Policy, on a home-built 3-DOF delta robot with an electromagnet gripper performing a bolt pick-and-place task. Both policies are trained on nested subsets of 10, 25, 50 and 100 teleoperated demonstrations for 5k, 20k and 50k optimizer steps, giving 24 models. Each model is scored on 10 real-robot trials, for 240 trials in total. Out of the box, LeRobot’s Diffusion Policy never left its start pose. Input-ablation tests showed that it had learned to copy the robot’s own joint state, a shortcut that leader–follower teleoperation data makes nearly optimal. An ImageNet-pretrained encoder, crop augmentation, a blinded joint-state input and a 1.6 s action horizon fixed it. With these changes the best model, Diffusion Policy with 100 demonstrations and 50k steps, succeeded on **10/10** trials; the best ACT models reached 6/10. Neither policy is usable with 10 demonstrations or 5k steps. ACT shows a clean data-scaling trend at 50k steps ($0 \rightarrow 20 \rightarrow 50 \rightarrow 60\%$, $p = 0.0015$). Pooled over the grid, the two policies are not significantly different (39/120 vs. 29/120, $p = 0.20$). They fail differently: ACT knocks the bolt over, Diffusion Policy stalls. We release the dataset, the trial log, all code, CAD and firmware, and the two best checkpoints.

1 Introduction

Imitation learning from teleoperated demonstrations has become the default way to teach low-cost robot arms manipulation skills. In open-source practice two policy classes dominate: ACT [1] and Diffusion Policy [2], both available in Hugging Face’s LeRobot library [3]. A practitioner adopting either one faces two immediate questions: *how many demonstrations are needed*, and *how long to train*.

Published comparisons mostly report one data and compute setting on standard serial arms. This study measures a full 4×3 grid of data and training length for both policies, on a mechanically different robot, a parallel (delta) robot with a binary magnetic gripper, and scores every model on the real robot under a fixed protocol.

Contributions.

- A controlled real-robot grid: 2 policies $\times$ 4 data sizes $\times$ 3 training lengths, 240 scored trials, released with the raw trial log and the code that regenerates every number and figure.
- A documented failure of default Diffusion Policy on leader–follower data, an instance of the “copycat” problem [4]; an offline diagnostic that detects it in minutes, and a fix.
- An open 100-episode, 3-camera dataset in LeRobot v3.0 format for a delta-robot embodiment, which is rare in public robot-learning data.
- Open hardware (STL and URDF), firmware, and recording and evaluation software.

2 Related Work

ACT [1] encodes camera images with a ResNet-18 [5] and predicts chunks of future actions with a conditional-VAE transformer. **Diffusion Policy** [2] predicts an action sequence by iterative denoising (DDPM [6]; DDIM [7] for faster sampling) conditioned on a short observation history. Both are implemented in LeRobot [3], which we use throughout.

Behaviour cloning is known to latch onto spurious, easily predicted signals. Causal confusion [8] and the copycat problem [4] describe policies that infer the next action from their own previous actions or state history instead of from the task-relevant observation, a form of shortcut learning [9]. We observe this effect concretely on real hardware with an off-the-shelf implementation, and show how a specific property of leader–follower data triggers it.

3 System

Robot. A 3-DOF delta robot. Three printed upper arms are driven by Feetech STS3215 serial-bus servos; steel parallelogram rods with ball joints carry the end effector. The gripper is an electromagnet (~ 12 V) switched through an H-bridge, which lifts a standing steel bolt. A Teensy 4.1 drives the servo bus at 1 Mbit/s. Joint commands are absolute angles in degrees, clamped to $[30^\circ, 110^\circ]$.

Teleoperation. Demonstrations are collected leader–follower (Fig. 1): a second, passive delta arm carries an

*Independent project, carried out outside of coursework and not supported or endorsed by Tufts University. I am a Tufts undergrad.



Figure 1: The two robots. **Left:** the servo-driven follower that the policies control, with the cup and a standing bolt. **Right:** the hand-moved leader arm used for teleoperation, with a magnetic encoder at each shoulder joint.

AS5600 magnetic encoder on each shoulder, and the Teensy drives the follower servos to the leader’s angles in real time. A push button switches the magnet. The Teensy streams leader angles, follower angles and magnet state at 60 Hz; the recorder samples at 30 Hz.

Cameras. Three Innomaker U20CAM-720P USB cameras (side, top_right, top_left, Fig. 2) are captured at 320×240 , 30 fps. Three uncompressed streams saturate a single USB hub, so capture forces MJPG through PyAV, and cameras are addressed by physical USB port rather than OS index. In all recorded data the side and top_right cameras have a constant yellow-green cast. Their auto white balance was off (4600 K) and exposure fixed at 2^{-6} s, a setting left behind by an earlier script and persisted by the Windows driver. For evaluation on Linux the same settings were reproduced through V4L2 controls and verified against the training video’s luma and chroma statistics.

4 Dataset

The dataset contains 100 demonstrations (33,200 frames at 30 Hz, 8.1–17.3 s each, median 10.7 s), recorded by one opera-



Figure 2: The policy’s input: one frame of demonstration 42 from the side, top_right and top_left cameras (320×240 each). Note the colour cast on the first two, constant across the dataset.

Table 1: Dataset summary (LeRobot v3.0, AV1 video, ~600 MB).

Episodes / frames	100 / 33,200 at 30 Hz
Observation	3 follower joint angles (deg); 3 RGB images 240×320
Action	3 leader joint angles (deg) + magnet $\{0, 1\}$
Task string	“pick up the bolt and drop it in the bucket”
Grasps	exactly one magnet press in 99/100 episodes
Start pose	$(91.7, 93.2, 90.8)^\circ$, s.d. $\approx 3.5^\circ$ per joint

tor on one day in ten sessions of ten. The observation is the follower’s three joint angles plus three RGB images; the action is the leader’s three joint angles plus the magnet command (Table 1).

Two properties matter for learning. First, every episode opens with ~ 1.2 s of stillness (median 1.23 s) and pauses ~ 0.33 s after the grasp. Second, because the action is the leader’s angle and the state is the follower’s, the action is always within a fraction of a degree of the next observed state. Section 6 shows why the second property matters.

Subsets. Training subsets are nested and seeded: the episodes are shuffled with a fixed seed and the first N are taken, so $\mathcal{D}_{10} \subset \mathcal{D}_{25} \subset \mathcal{D}_{50} \subset \mathcal{D}_{100}$ (3,219 / 8,380 / 16,326 / 33,200 frames).

5 Training

All models use LeRobot 0.5.0, PyTorch 2.6 in fp32, batch size 32, seed 1000 and a single RTX 3090. Batch size 32 was the largest at which Diffusion Policy fits in 24 GB. Passes over the data range from 5 epochs (100 demos, 5k steps) to 497 (10 demos, 50k steps).

ACT. LeRobot defaults: ImageNet-pretrained ResNet-18, one observation frame, chunk size 100 with all 100 actions executed before re-planning, VAE with KL weight 10, and a constant learning rate of 10^{-5} . Because the learning rate is constant, one 50k-step run per data size supplies the 5k and 20k cells as intermediate checkpoints.

Diffusion Policy. The cosine learning-rate schedule depends on the total number of steps, so each of the 12 cells is a separate run. The LeRobot defaults (“v1”) did not work on the robot (Section 6). All scored Diffusion results use the “v2” configuration in Table 2; everything else (subsets, batch size, seed, schedule) matches v1.

Table 2: Diffusion Policy: LeRobot defaults (v1) vs. the configuration used for all scored results (v2). The 48/24 horizon reproduces the original paper’s 1.6 s / 0.8 s at 30 Hz.

	v1 (default)	v2
Vision encoder	ResNet-18, scratch, GroupNorm	ImageNet, BatchNorm
Augmentation	none	random crop 216×288
Observation frames	2	1
Joint state	used	zeroed (train + test)
Horizon / executed	16 / 8 (0.53 s)	48 / 24 (1.6 s)

Training loss. All 28 grid runs completed and their loss decreased throughout (Fig. 3). For both policies, *fewer* demonstrations gave *lower* final loss (memorization), the opposite of the robot-success ordering. ACT’s loss (L1 + KL) and Diffusion’s (noise-prediction MSE) are not comparable with each other.

A LeRobot 0.5.0 bug. With episode subsets, Diffusion’s episode-aware sampler yields absolute frame indices into a dataset indexed relative to the subset. Training then crashes or silently reads the wrong frames. A small wrapper that remaps the indices (released) was used for every run.

6 The Copycat Failure of Default Diffusion Policy

Symptom. On the real robot every v1 Diffusion checkpoint held still at its start pose. Changing the inference setup did not help: GPU or CPU, DDPM-100 or DDIM-10, fixed initial noise, longer executed chunks, or a scripted “kick-start” lift, after which the arm stopped again. ACT on the same rig moved normally.

Diagnosis. We probed checkpoints offline on training episodes:

- *Teacher-forced replay* looked perfect ($\sim 0.4^\circ$ error), so it reveals nothing.
- *Images only* (images play the demonstration, joint state frozen at frame 0): ACT reproduced 92% of the demonstrated motion range, v1 Diffusion only 37%.
- *State only* (images frozen, state moving): v1 Diffusion followed the state.
- *Still start* (robot at rest, first camera frame): v1 Diffusion (50 demos, 50k steps) commanded $< 1^\circ$ of motion, and longer training made this worse.

The policy had learned “next action $\approx$ current joint state (+ velocity from the two observation frames)”. With leader-follower data this is a near-perfect predictor over a short horizon, and a still robot then predicts a still robot. The v1 defaults favour this shortcut: a vision encoder trained from scratch, no augmentation, two state frames that expose velocity, and a 0.53 s horizon. ACT’s pretrained encoder, single frame and 3.3 s chunks push it towards the images.

Table 3: Successes out of 10 trials per model. Superscripts: trials stopped as unsafe.

Demos	ACT			Diffusion Policy (v2)		
	5k	20k	50k	5k	20k	50k
10	0 ¹⁰	0 ²	0	0 ¹⁰	0 ²	2
25	0 ²	6	2	1 ²	8	5
50	1 ⁴	6 ¹	5	2 ²	4	1
100	0 ¹⁰	3	6	0 ²	6	10
Total	29/120 (24%)			39/120 (33%)		

Fix. Figure 4 shows the pilot variants (50 demos, 5k steps). Pretrained vision with crop augmentation barely helped (42%). *Blinding the joint state* was decisive (96%): the policy moved and grasped, but jittered and hesitated. Extending the horizon to 1.6 s removed most of the jitter, and this v2 pilot completed a pick-and-place. Fully trained v2 models were 99–100% image-driven by this measure.

7 Robot Evaluation

Protocol. Five bolt positions are marked on the table. P1–P3 lie inside the demonstrated region. P4 lies inside it but is partly hidden by the cup and awkward to reach. P5 lies outside the demonstrated region but is easy to reach. Each model receives two trials per position (10 per model). The cup position, lighting and camera setup are fixed, and models are tested in a seeded random order. Each trial starts from the dataset’s mean start pose and runs at 30 Hz until the operator stops it or 900 actions (30 s of motion) elapse. Success means the bolt ends in the cup. The operator also logged whether the bolt was picked up, the number of grasp attempts and a failure mode. Unsafe stops, where the operator halts erratic motion, count as failures.

Inference. ACT ran on a laptop CPU (Ryzen AI 9 HX 370, 0.1–0.37 s per 100-action chunk, 29.3–29.9 Hz measured). Diffusion v2 ran on the RTX 3090 with DDPM-100 as trained, taking ~ 0.76 s per 24-action plan; the robot pauses while it plans, which is acceptable for this quasi-static task. On the laptop CPU, Diffusion would take ~ 18 s per plan.

7.1 Results

Table 3 and Fig. 5 give the full grid; Figs. 6 and 7 show the two scaling axes. Table 4 lists the statistical tests, with trends tested by Cochran–Armitage [11] on $\log_2$ of the varied quantity and policy comparisons by Fisher’s exact test.

Clustering caveat. The two trials at the same position agreed in 88% of pairs, so each model’s 10 trials carry roughly the information of 5 independent outcomes. All p -values should therefore be read as optimistic, and single cells as noisy: a 6/10 cell has a 95% interval of 31–83%.

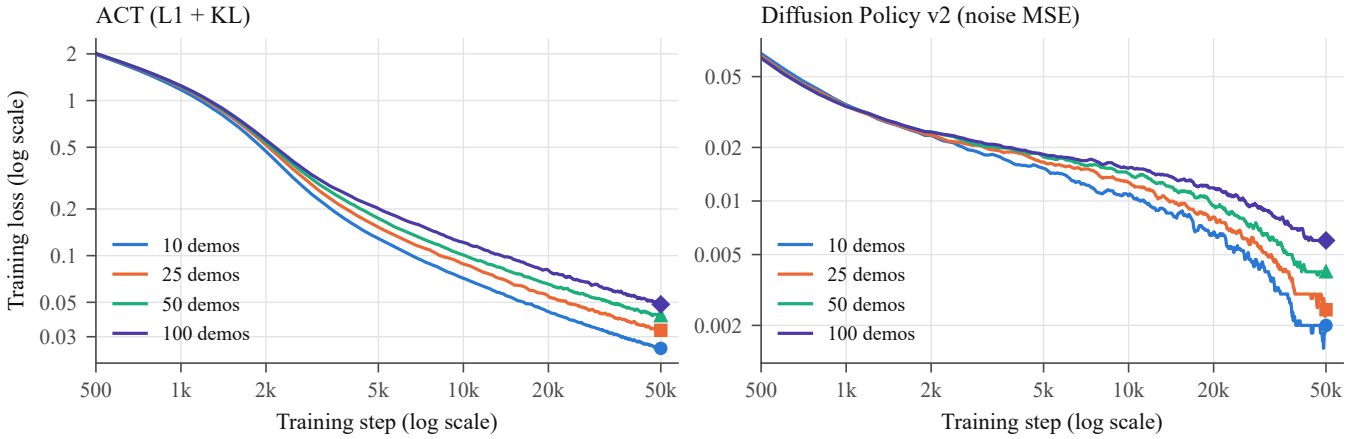


Figure 3: Training loss of the 50k-step runs from step 500 onward (lightly smoothed, log-log axes). Fewer demonstrations always reach a lower loss, while robot success moves the other way (Fig. 5). Losses are not comparable across the two panels.

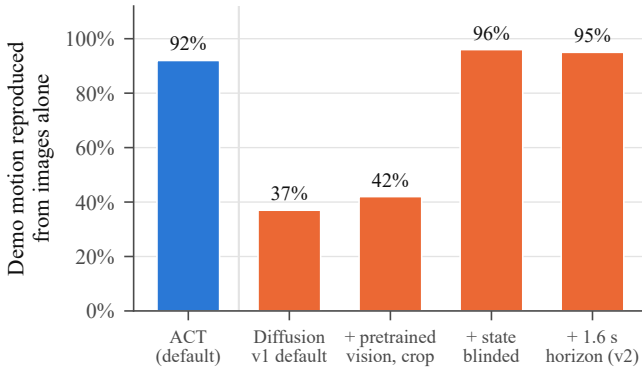


Figure 4: Images-only ablation: the share of the demonstrated motion range a policy reproduces when the joint state is frozen and only the images move. Diffusion bars are cumulative changes to the LeRobot defaults (50 demos, 5k steps).

Table 4: Statistical tests (two-sided).

Comparison	Successes	p
ACT, demos trend @ 50k	0, 2, 5, 6	0.0015
ACT, steps trend @ 100 demos	0, 3, 6	0.004
Diffusion, steps trend @ 100 demos	0, 6, 10	$< 10^{-4}$
Diffusion, demos trend @ 50k	2, 5, 1, 10	0.005 [†]
ACT vs. Diffusion, pooled (12 cells)	29 vs. 39 / 120	0.20
ACT vs. Diffusion, 100 demos @ 50k	6 vs. 10 / 10	0.087

[†] significant but not monotone.

Failure modes. The policies fail in characteristically different ways (Fig. 8). ACT fails *aggressively*: it drives to the bolt and grasps imprecisely, knocking the standing bolt over (39 trials). Undertrained ACT at 5k steps was erratic enough to be stopped on every trial at both 10 and 100 demonstrations. Diffusion v2 fails *hesitantly*: it hovers or freezes (52 stalls), most often before the grasp. This matches the still segments in the demonstrations. With the state blinded, a still camera view is ambiguous between “about to move” and “waiting”, and an undertrained model can stay in the waiting mode.

Bolt position. P4, the occluded position, was the hardest (Fig. 9): ACT succeeded once in 24 trials and Diffusion 7 times, and P4 accounted for 21 of the 47 unsafe stops. The out-of-distribution position P5 was *not* the hardest: both 100-demo ACT models and the best Diffusion model went 2/2 there. Unexpectedly, both 100-demo ACT models failed at P1, the most heavily demonstrated position, while the 50-demo ACT models succeeded there. One untested explanation is that ACT’s latent is fixed to its mean at test time, so many slightly different P1 grasps average into one a few millimetres off, and the magnet needs near-contact.

Best model. Diffusion v2 with 100 demonstrations and 50k steps succeeded on all 10 trials, at every position including the occluded P4 and out-of-distribution P5, each with a single grasp attempt (Fig. 10). Subjectively it was also the most convincing policy in the study. Three ACT models reached 6/10. We release the 100-demo, 50k-step ACT as the reference ACT model because it runs in real time on a laptop CPU and was never stopped as unsafe.

8 Discussion

Training length is the first gate. At 5k steps neither policy is usable (4/80 trials combined), whatever the data size; for 100 demonstrations this is only 5 epochs. Given enough steps, more data helps both policies, most clearly at 50k.

Possible overtraining at small and medium data. At 25 demonstrations both policies did better at 20k than at 50k steps (ACT 6 $\rightarrow$ 2, Diffusion 8 $\rightarrow$ 5), and Diffusion at 50 demonstrations fell from 4 to 1. No single drop is significant at $n = 10$, but the direction repeats three times while training loss keeps falling, and at 100 demonstrations longer training only helped. Scaling training steps with dataset size, rather than training small datasets longer, is a reasonable hypothesis that this data cannot confirm.

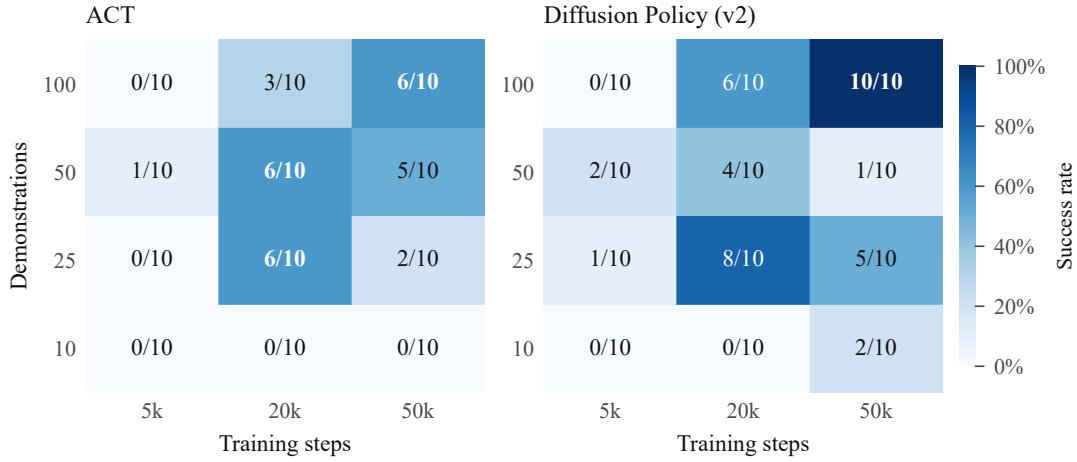


Figure 5: Real-robot success for every model (10 trials each).

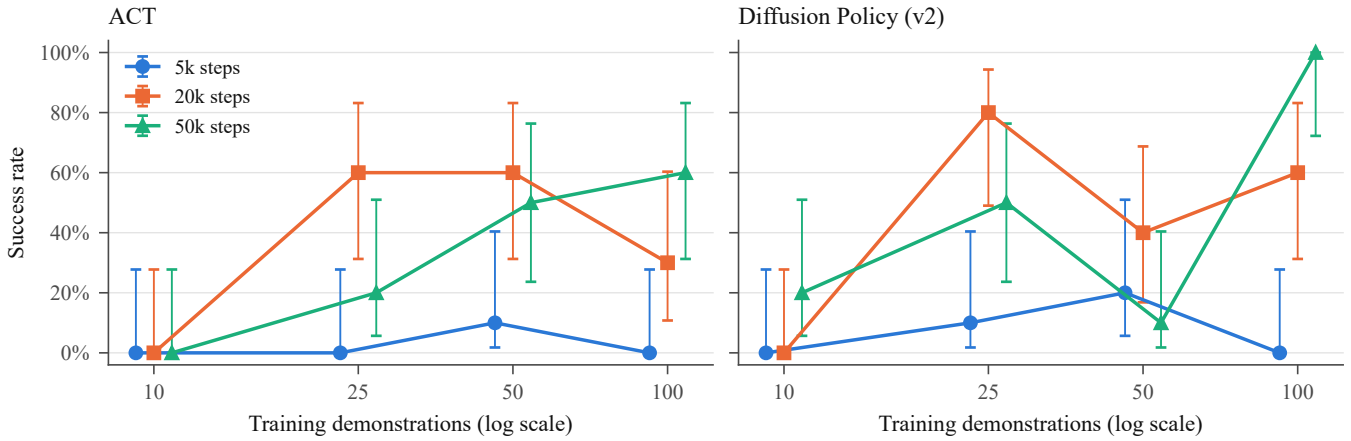


Figure 6: Success rate against the number of demonstrations, one line per training length. Error bars are 95% Wilson intervals [10] ($n = 10$); points are dodged horizontally for legibility.

Policy choice is a trade-off. Modified Diffusion Policy produced the single best model and handled the occluded position far better. ACT was competitive in the middle of the grid, runs in real time on a laptop CPU, and needed no changes. The pooled difference is within noise.

Proprioception and leader-follower data. The same data trained ACT without trouble, but ACT’s loss of precision at P1 with 100 demonstrations hints that it may also lean on state. A proprioception ablation for ACT is the natural next experiment. Diffusion’s stalling probably comes from the still segments in the data, so trimming them is a likely remedy. For dataset builders the lesson is that leader-follower data invites state-copying, and that low loss or good teacher-forced replay does not rule it out, whereas an images-only ablation does.

9 Limitations

- **Unequal conditions.** ACT was evaluated on a Windows laptop CPU on 2–3 October and Diffusion on a Linux GPU desktop on 5 October. The random order held within each policy, not across them, so drift between days could align

with policy.

- **Diffusion v2 is not stock Diffusion Policy:** it has no proprioception, pretrained vision, crop augmentation and a longer horizon, whereas ACT used defaults. Stock v1 was not formally scored; it never completed a pick in informal testing.
- **Small samples:** 10 trials per model, 5 positions, strong within-position clustering, and one training seed per cell.
- **Non-blind scoring:** the operator knew which model was running.
- **Hardware events during ACT testing:** a printed upper arm was replaced with an identical part; a faulty leader-encoder cable was unplugged; the firmware was patched to skip the missing encoders in autonomous mode; a loose magnet wire was fixed. All scored ACT runs logged ~ 29.5 Hz control.
- **Scene drift** between recording and testing was corrected by restoring the scene against the demonstration videos.

10 Conclusion

On a home-built delta robot doing pick-and-place, both ACT and Diffusion Policy need roughly ≥ 25 demonstrations and

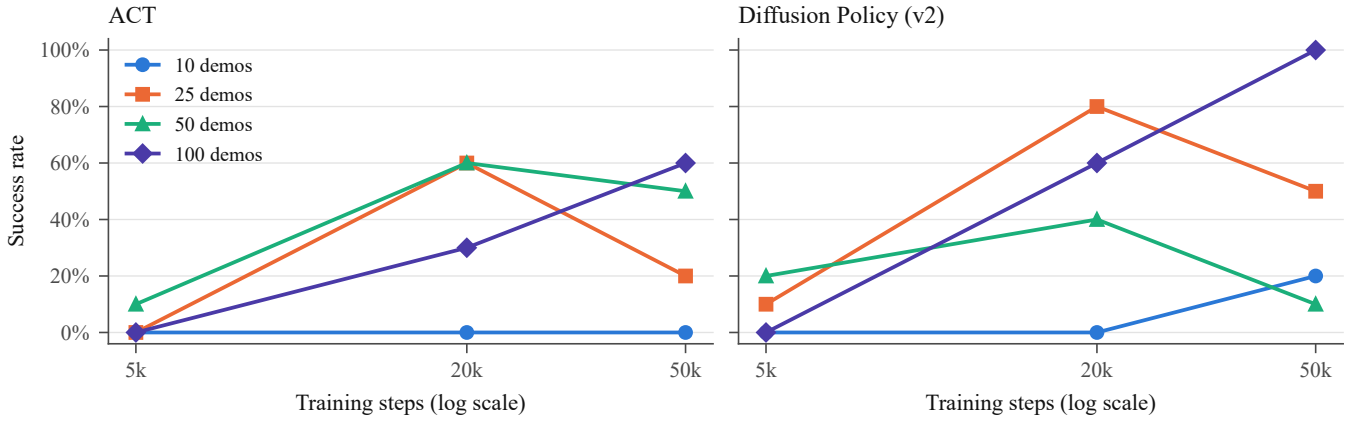


Figure 7: Success rate against training length, one line per data size.

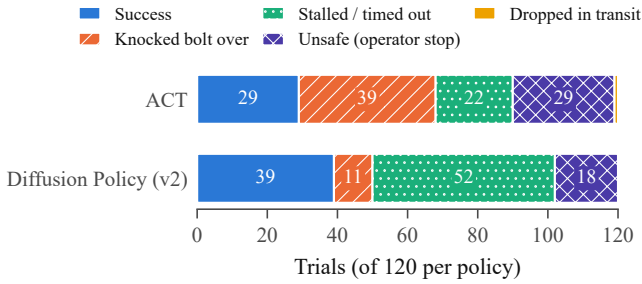


Figure 8: Outcome of all 120 trials per policy.

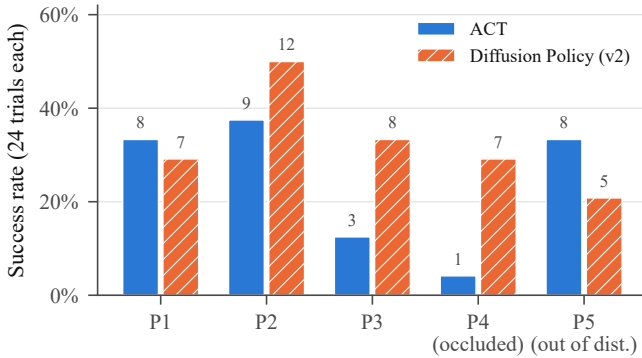


Figure 9: Success by bolt position, pooled over each policy’s 12 models (numbers: successes out of 24).

$\geq 20k$ training steps before they work at all, and more data pays off most with long training. The two policies fail in characteristically different ways. Default Diffusion Policy can silently learn to ignore its cameras on leader–follower data; a simple ablation exposes this and blinding proprioception fixes it, after which Diffusion Policy trained on 100 demonstrations for 50k steps solved the task in 10 of 10 trials. Future work: more trials and seeds with blind scoring, a proprioception ablation for ACT, trimming still frames from the demonstrations, temporal ensembling for ACT, and faster Diffusion sampling for real-time control.

Release. Dataset, models: <https://huggingface.co/KirkHaskell>. Code, CAD, firmware, trial log and

figure scripts: <https://github.com/KirkEHaskell/delta-robot-act-vs-diffusion>.

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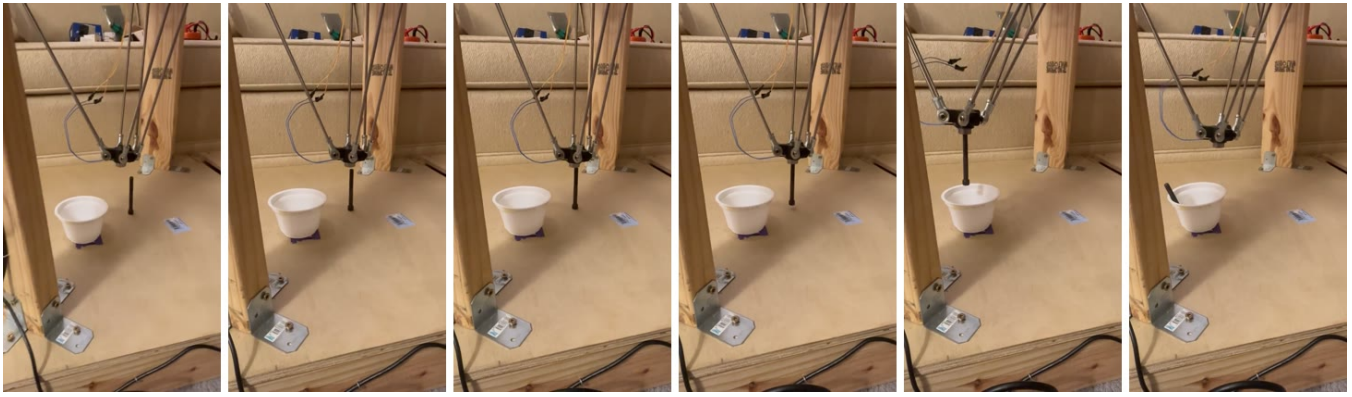


Figure 10: The best model (Diffusion Policy v2, 100 demonstrations, 50k steps) completing the P3 task autonomously: approach, magnetic grasp, transport, and release into the cup (frames ~ 2 s apart).

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